

How Does Cooperation Reduce Intergroup Bias?

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This experiment examined the hypothesis derived from the social categorization perspective that intergroup cooperation reduces bias by transforming members' cognitive representations of the aggregate from 2 groups to 1 group. Two 3-person groups experienced intergroup contact under conditions that varied (a) members' representations of the aggregate as 1 group or 2 groups (without involving cooperation) and (b) the presence or absence of intergroup cooperation. As expected, in the absence of cooperation, bias was lower among Ss induced to conceive of the 6 participants as 1 group rather than as 2 groups. Also as predicted, among Ss in the 2-groups condition, intergroup cooperation increased the strength of the 1-group representation and decreased bias. Multiple regression mediation analysis revealed, as expected, that members' representations mediated bias and that the 1-group representation primarily increased the attractiveness of former outgroup members.

Despite substantial documentation that intergroup cooperative interaction reduces bias (Allport, 1954; Aronson, Blaney, Stephan, Sikes, & Snapp, 1978; Cook, 1985; Deutsch, 1973; Johnson, Johnson, & Maruyama, 1983; Sherif, Harvey, White, Hood, & Sherif, 1954; Slavin, 1985; Worchel, 1979), it is not clear how cooperation achieves this effect. One basic issue involves the psychological processes that mediate this change. A second issue concerns which specific aspects of intergroup cooperation (e.g., interaction, common goals, and common fate), as well as what other general features of the contact situation (e.g., equal status between the groups, equalitarian norms, opportunities to learn stereotype-inconsistent information; see Cook, 1985), are necessary and sufficient to reduce intergroup bias. The present research addressed primarily the first issue: How does cooperative interaction reduce bias?

The classic functional relations perspective by Sherif et al. (1954) views cooperative interdependence as a direct mediator of attitudinal and behavioral changes. However, several addi-

tional explanations have been proposed (see Brewer & Miller, 1984; Miller & Davidson-Podgorny, 1987; Worchel 1979, 1986). For example, cooperation may induce greater intergroup acceptance as a result of dissonance reduction serving to justify this type of interaction with the other group; as a result of the positive, reinforcing consequences of mutual cooperation (Lott & Lott, 1965); or as the result of increases in knowledge about the other group, which reduces intergroup anxiety (Stephan & Stephan, 1984). An additional model that may account for the effects of cooperation is based on Tajfel's (1969) and Tajfel and Turner's (1979) work on social categorization and social identity theory. This model proposes that intergroup cooperation promotes intergroup acceptance because it reduces the cognitive salience of the intergroup boundary (see also Wilder, 1986). A recent position proposed by Brewer and Miller is aligned with the categorization model; this position proposes in addition that when intergroup cooperative interaction permits members' attention to focus on one another's personal qualities, it contributes to the development of personalized rather than categorized interactions and thereby to reduced bias (see also Miller & Brewer, 1986; Miller, Brewer, & Edwards, 1985). Also, Neuberg and Fiske (1987) have shown more generally that cooperative interdependence increases motivation to form more individuated impressions.

Within the framework of the social categorization model, the present study examined the hypothesis that intergroup cooperation reduces bias, at least partially, because intergroup coopera-

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tion reduces the salience of the intergroup boundary. Specifically we proposed that intergroup cooperation induces the members to conceive of themselves as one (superordinate) group rather than as two separate groups, thereby transforming their categorized representations from *us* and *them* to a more inclusive *we* (see Brown & Turner, 1981; Campbell, 1958; Doise, 1978; Feshbach & Singer, 1957; Hornstein, 1976; Turner, 1981; Worchel, Axson, Ferris, Samaha, & Schweitzer, 1978). Indeed, Sherif and Sherif (1969, pp. 268–269) acknowledged the potential that intergroup cooperation has toward facilitating the development of a common superordinate entity. This possibility, however, was conceived by Sherif and Sherif to represent the very gradual development of a highly structured superordinate group, rather than the immediate creation of a social entity that may only exist more ephemerally within the perceptions of one or more of its members.

Assuming that intergroup cooperation may initially transform members' cognitive representations from two groups to one group, how might this change reduce bias? The process by which a revised one-group representation could reduce intergroup bias rests theoretically on two related conclusions of Brewer's (1979) analysis of the origins of intergroup bias and on propositions underlying social identity theory (Tajfel & Turner, 1979) and self-categorization theory (Turner, 1985; Turner, Hogg, Oakes, Reicher, & Wetherell, 1987; see also Hogg & Abrams, 1988). First, intergroup bias frequently takes the form of in-group enhancement rather than out-group devaluation (see Rosenbaum & Holtz, 1985, for a different perspective). Second, group formation brings in-group members closer to the self, but the distance between the self and out-group (or non-in-group) members remains relatively unchanged. Consequently, circumstances that induce the memberships to adopt a one-group representation can extend the cognitive and motivational processes that brought original in-group members closer to the self and thereby enhance attitudes toward the former out-group members.

This reasoning is supported in an earlier study (Gaertner, Mann, Murrell, & Dovidio, 1989), in which the members of two 3-person groups were induced to conceive of the total 6-person aggregate as one group, two groups, or as six separate individuals (i.e., no groups). As expected, the one-group and the separate-individuals conditions each had lower levels of intergroup bias than did the two-groups control condition (which maintained the salience of the intergroup boundary). Furthermore, supportive of Brewer's (1979) analysis as well as self-categorization theory (Turner, 1985; Turner et al., 1987), these recategorized conditions reduced bias in different ways. In the one-group condition, bias was reduced primarily by increasing the attractiveness of former out-group members, whereas in the separate-individuals condition, bias was reduced primarily by decreasing the attractiveness of former in-group members. Thus, as specified by Turner et al.'s self-categorization theory, "the attractiveness of an individual is not constant, but varies with in-group membership" (p. 60). Consequently, to the extent that the proposed changes in members' conceptual representations from two groups to one group mediate the causal relation between intergroup cooperation and reduced intergroup bias,

bias should diminish primarily because of the development of more positive attitudes toward former out-group members.

The present study was designed to assess the efficacy of the proposed recategorization model. Most likely, intergroup cooperation reduces bias by means of several different pathways, only one of which involves transforming the memberships' representations of the aggregate. By tracking the status of subjects' cognitive representations of the memberships in the present study, we investigated whether the proposed cognitive changes played a role in reducing intergroup bias.

To examine the hypothesis that the causal relation between intergroup cooperation and reduced intergroup bias is mediated (at least partially) by members' transformed representations of the aggregate from two groups to one group, we initially created two separate 3-person laboratory groups and then brought these groups into contact under circumstances designed to vary (a) the members' representations of the aggregate as either one group or two groups and (b) the presence or absence of intergroup cooperative interaction.

Members' representations of the aggregate were manipulated by exposing them to a set of systematically varied features (not involving cooperation), which in concert were expected to influence their conceptual representations of the memberships as one group or two groups (e.g., see Gaertner et al., 1989). For example, we varied whether the two groups sat at a single table in an alternating seating pattern or at two separate tables (see also Ryen 1974) and whether they were assigned a new single name to represent both groups or maintained their earlier selected, separate group names, as well as other features.

Intergroup cooperation is a complex process that usually involves several features, including interaction, common problems or goals, and common fate between the memberships. Brewer's (1979) review noted that the introduction of a cooperative intergroup reward structure alone, without intergroup interaction, is not sufficient to eliminate intergroup bias. Also, when intergroup cooperation has been operationalized in natural settings (see Johnson et al., 1983; Sherif et al., 1954; and Slavin, 1985), it specifically involves the opportunity for interaction, common goals, and common fate between the memberships. Therefore, to manipulate intergroup cooperation in the current experiment, we varied the presence or absence of all three features simultaneously. In the cooperation condition, there was an opportunity for interaction between the memberships during the joint development of a single solution to a common problem while the groups shared a common fate that turned on the effectiveness of their joint solution. In the no-cooperation condition, both groups together (in either the one-group or two-groups conditions) listened to a tape recording of some other group's discussion and then, as individuals and without interdependence, rated the effectiveness of the recorded group's discussion.

Although an assessment of which specific features of intergroup cooperation are necessary and sufficient to reduce bias would be an important undertaking, this was not the goal of the current study. Rather, in the context of varying (in an all or none manner) the three major features of intergroup cooperation, this study began to address our primary question, which involved the psychological processes that mediate the relation-

ship between intergroup cooperation and reduced bias. The no-cooperation/two-groups condition (which emphasized the two-groups categorization) is especially pertinent because of its partial structural similarity to Tajfel, Billig, Bundy, and Flament's (1971) minimal intergroup situation, except that in the present study all participants met face to face, following an initial period of in-group interaction. In Tajfel et al.'s work, members were isolated from each other, and intergroup bias resulted exclusively from the simple categorization of two groups. Tajfel et al.'s two groups, like our own, existed in the absence of explicit competition, interaction, a common problem, or common fate between the memberships. Thus, with circumstances similar in these ways to Tajfel et al.'s to serve as the no-cooperation baseline, the current study examined the consequences of introducing the major features of intergroup cooperation. The dependent variables included measures of subjects' representations of the aggregate (e.g., "Does it [the aggregate] feel like one group, two groups or separate individuals?"), of how cooperative were all 6 participants, and of evaluative ratings of in-group and out-group members.

The experimental design and the results of two comparisons permits a test of the hypothesis that a one-group representation mediates the causal relation between intergroup cooperation and reduced intergroup bias. In particular, one assumption of the present research was that there is a causal link between a one-group representation and reduced bias, even in the absence of cooperative interaction. Thus, within only the no-cooperation conditions of the present study, bias should be lower in the one-group than in the two-groups condition. In addition, if cooperation transforms members' representations from two groups to one group, then within only the two-groups representation conditions comparisons involving the no-cooperation and cooperation conditions should indicate that the introduction of intergroup cooperation results in an increase in the extent to which the aggregate feels like one group (e.g., an increase on a scale ranging from 1 to 7 that measured how much the aggregate feels like one group) and a decrease in intergroup bias. Furthermore, we predicted that this reduction in bias, which is mediated by changes in members' representations, would occur primarily because attitudes toward former out-group members become more positive. In addition, the major hypothesis that a one-group representation mediates the causal relation between intergroup cooperation and reduced intergroup bias was evaluated with the multiple regression mediation approach suggested by Baron and Kenny (1986) and Judd and Kenny (1981).

Method

Subjects

Four hundred seventy-four students (234 men and 240 women) enrolled in the general psychology course participated in partial fulfillment of their research-readings or participation requirement. Testing the subjects in same-sex groups of 6 people per session, our intention was to assign 10 groups of men and 10 groups of women to each of the 4 experimental cells. In error, only 9 groups of men were assigned to the cooperation/one-group cell; therefore only 79 six-person groups were run.

Procedure

Subgroup formation. In each session, two 3-person groups were assigned ostensibly to two different experiments located in different areas within the laboratory complex. Each group was unaware of the other group's existence. Greeted by different experimenters, members of each group were assigned one of three different color-coded identity tags reserved for each person (purple, yellow, and brown; or green, orange, and red) as they were led to their designated laboratory. As they entered, the experimenter asked each person to attach these identification tags to their clothing and to sit according to their color-coded identity, which was matched by a color-coded placemat on the group's table. Each room was also equipped with a visible video camera and microphone. Tape-recorded instructions explained that the study involved the examination of group decision-making processes and that we would be videotaping their group interaction as they attempted to reach consensus to the winter survival problem (Johnson & Johnson, 1975). Members of each group were first asked to create a name for their group and to record it on their special group consensus form as well as on their personal identity tags. Following the 3-person group discussion focused on the winter survival problem, subjects were informed of their subsequent contact with the other group, which would take place soon in a larger room within the laboratory complex.

Cooperation manipulation. Overall, three aspects of intergroup cooperation were varied simultaneously: interaction, a common problem requiring a consensus solution, and common fate. In the no-cooperation condition, none of these features were present. Instead, the members of both groups were brought together and exposed to a tape-recorded discussion of two other groups reaching consensus on the winter survival problem. Following exposure to the 6-min recording, each participant recorded an evaluation of the effectiveness of the prerecorded group's solution. In the cooperation condition, all three features of cooperation were present. Specifically, the two groups were interdependent in that they discussed the winter survival problem and reached a single consensus solution that if effective (in terms of an absolute standard determined by the experimenters) would qualify the 6 participants for entry into a lottery at the end of the semester during which they could share a \$60 prize (i.e., \$10 per person). No feedback about the group's effectiveness was given during the experimental session.

Representation manipulation (one group vs. two groups). The goal of the representation manipulation was to vary the subjects' cognitive representations of the 6 participants as either one group or as two separate groups; it was very similar operationally to Gaertner et al. (1989). To induce these groups to have a one-group representation of the aggregate during contact (independent of the presence or absence of cooperative interaction), the members of both groups were seated in same colored chairs at a single hexagonal table in an integrated seating pattern (ABABAB). Subsequent instructions emphasized their common identity as students at the same university and instructed them all to wear the same available T-shirts displaying the name of their university. Also, an experimenter assigned the members of both groups a new single name to represent all 6 people and instructed them to replace the name of their original 3-person group written on their personal identity tags with this new name. The experimenter announced, ostensibly for the TV recording, that the _____ (3-person group name) and the _____ (3-person group name) had merged to form the STARS (new 6-person group name), which members were also asked to use on all subsequent experimental forms. Also, the two visible TV cameras were each directed at all 6 people during the contact period. In addition, all participants were asked to hold on to a rope placed in the center of the table, ostensibly to keep all participants within camera range. Last, because in the cooperation conditions, the derived consensus solution

would function as a tangible product to represent all 6 participants, subjects in the no-cooperation one-group condition were instructed to estimate (based on observations of others' nonverbal reactions) the 6 current participants' average evaluative rating of the prerecorded group's solution. Thus, each subject in the no-cooperation one-group condition derived a single rating to represent all 6 participants but without cooperative interaction or interdependence as in the cooperation/one-group condition.

In the two-groups condition, the members of each 3-person group sat at different tables (36 in. apart) in a segregated seating pattern (AAABBB), in different colored chairs (one group used red chairs; the other group used yellow chairs), during exposure to instructions that did not mention their common university identity or require them to wear university T-shirts. Also, each group maintained their separate 3-person group names on their identity tags throughout, and the experimenter announced the session using the names of both groups without mention of any merger. Also, in the two-groups condition, the 6 participants did not hold on to a rope, nor did they estimate an average of the other participants' ratings of the group's interaction. Instead, each participant in the no-cooperation/two-groups condition separately recorded his or her own evaluation of the effectiveness of the prerecorded group's discussion. Finally, each of the two TV cameras was aimed at a different 3-person group throughout.

Measures. Following the experimental manipulations, the participants were given a postexperimental questionnaire containing items which asked about (a) their conceptual representations of the aggregate ("Which description best characterizes your impression of all of you currently participating in this experiment? It felt most like one group, two groups, or separate individuals"), as well as to what extent, *not at all* (1) to *very much* (7), it felt like each of these representations (after Gaertner et al., 1989); (b) their ratings of how much it felt like they were cooperating and competing with each other during the contact period; and (c) their evaluative ratings (1–7) of each participant (except for themselves) regarding how much they liked each of the others, as well as their ratings of each person's honesty, cooperativeness, and similarity to themselves. Separate in-group and out-group ratings were constructed for each person and then averaged for each 6-person group. Thus ratings of in-group and out-group members were based on ratings by individuals of individuals from which group-level evaluations were constructed.

Results

Because of the possible interdependence among the 6 participants per session, the unit of analysis throughout all statistical analyses was the group mean for each 6-person group ($n = 79$) rather than each individual's ratings.

Efficacy of the Experimental Manipulations

When asked to select the representation (one group, two groups, or separate individuals) that best characterized their impressions of the aggregate, sizable percentages of subjects within each 6-person group in the one-group and two-groups conditions selected the alternative most appropriate to their treatment condition (see Table 1). Separate 2 (representation) $\times$ 2 (cooperation) $\times$ 2 (sex of subject) analyses of variance (ANOVAs) conducted on the average percentage of subjects per group ($n =$

Table 1
Effects of Cooperation and Representation on Members' Impressions of the Interaction

Impression	No cooperation		Cooperation	
	Two groups ($n = 20$)	One group ($n = 20$)	Two groups ($n = 20$)	One group ($n = 19$)
Members' representations				
Percentage				
One group	15.8 _a	39.7 _b	48.3 _{bc}	66.7 _c
Two groups	75.0 _a	47.8 _b	47.5 _{bc}	28.9 _c
Separate individuals	9.0	12.5	4.2	4.4
Mean rating (1–7)				
One group	2.7 _a	3.9 _b	4.4 _{bc}	5.1 _c
Two groups	5.1 _a	3.5 _b	3.3 _b	2.3 _c
Separate individuals	2.8 _a	2.5 _a	1.7 _b	1.7 _b
We were cooperating	4.6 _a	4.7 _a	5.8 _b	5.9 _b
We were competing	2.2 _a	2.2 _a	2.2 _a	1.7 _b

Note. Means with different subscripts within a row are reliably ($p < .05$) different than one another according to Tukey's honestly significant difference method.

79) selecting the one-group and two-groups alternatives each yielded main effects for the representation and cooperation manipulations and no effect for sex of subject. Supportive of the efficacy of the representation manipulation, greater percentages of subjects in the one-group condition (52.8%) selected the one-group alternative in relation to those in the two-groups condition (32.1%), $F(1, 71) = 16.03$, $p < .001$. Correspondingly, greater percentages of participants in the two-groups condition selected the two-groups alternative (61.25%) than did subjects in the one-group condition (38.63%), $F(1, 71) = 21.40$, $p < .001$. In addition, the efficacy of the representation manipulation was supported by similar ANOVAs conducted on subjects' separate ratings of the extent (1–7) to which the aggregate felt like one group and two groups. As indicated in Table 1, subjects in the one-group, in relation to the two-groups, condition saw the aggregate more as one group, $F(1, 71) = 26.30$, $p < .001$, and less as two groups, $F(1, 71) = 67.98$, $p < .001$.

The efficacy of the cooperation manipulation was evaluated by additional 2 (representation) $\times$ 2 (cooperation) $\times$ 2 (sex) ANOVAs on subjects' ratings (1–7) of the levels of cooperation during the contact period. As expected, subjects in the cooperation condition rated the participants as more cooperative ($M = 5.86$) than did participants in the no-cooperation condition ($M = 4.63$, $F(1, 71) = 139.25$, $p < .001$). Parallel but substantially weaker effects were obtained for subjects' ratings of competition within the cooperation ($M = 1.98$) and no-cooperation ($M = 2.20$) conditions, $F(1, 71) = 4.34$, $p < .04$. Thus, subjects' ratings of the contact period supported the efficacy of the representation and cooperation manipulations.

Does Cooperation Transform Members' Representations?

If subjects' representations of the aggregate mediate the causal relationship between intergroup cooperation and intergroup bias, then each of the ANOVAs involving the measures of

subjects' representations should have also yielded main effects for the manipulation of cooperation. Indeed, in addition to the effects for representation, each of the analyses reported earlier did obtain main effects for cooperation. For example, supportive of the idea that intergroup cooperation transforms members' conceptual representations of the aggregate from two groups to one group, the separate ANOVA involving the average percentages of subjects selecting the one-group alternative (as opposed to two groups or separate individuals) yielded a main effect for cooperation, $F(1, 71) = 31.40, p < .001$. As expected, when the groups cooperated, higher percentages of subjects indicated that the aggregate felt like one group (57%) in relation to when the opportunity for cooperation was not available (28%). Corresponding effects were obtained for the percentages of subjects selecting the two-groups alternative, $F(1, 71) = 21.44, p < .001$. Also, main effects for cooperation were obtained on the (1-7) measures of the extent to which the aggregate felt like one group, $F(1, 71) = 67.66, p < .001$, and two groups, $F(1, 71) = 93.90, p < .001$. As indicated in Table 1, subjects in the cooperation condition, in relation to the no-cooperation condition, saw the aggregate more as one group and less as two groups.

Although not specifically expected, main effects for cooperation also were obtained for the percentages of subjects selecting the separate-individuals alternative, $F(1, 72) = 4.39, p < .04$, and for subjects' ratings of the extent (1-7) to which the aggregate felt like separate individuals, $F(1, 71) = 29.34, p < .001$. Within the cooperation condition, somewhat smaller percentages of subjects selected the separate-individuals representation and rated it more weakly than did subjects in the no-cooperation condition (4.3% vs. 10.8%, respectively, and 1.71 vs. 2.63). Thus, although the cooperation manipulation affected subjects' impressions regarding the separate-individuals possibility, this particular representation was less common and weaker than the alternative representations (see Table 1).

How Does Cooperation Reduce Intergroup Bias?

The measures of intergroup bias involved subjects' ratings (1-7) of in-group and out-group members' likability, cooperativeness, honesty, and similarity to the self. The internal consistency for these ratings by each group was high, as indicated by Cronbach's alpha (in-group = .89; out-group = .93). Multivariate comparisons involving these four ratings were conducted in accordance with the predictions derived from the hypothesis that a one-group representation would mediate the causal relation between intergroup cooperation and reduced bias. Separate indexes for in-group and out-group members consisting of the mean of these four evaluative ratings (Table 2) also were calculated to illustrate more clearly the pattern of findings associated with the multivariate effects.¹

A major assumption of the present research was that there is a causal link between a one-group representation and reduced bias, even in the absence of cooperative interaction. Thus, considering only the no-cooperation conditions of the present study ($n = 40$ groups), if subjects' representations of the aggregate were transformed as expected, bias would be lower in the one-group than in the two-groups condition.

First, as expected within the no-cooperation conditions, sub-

Table 2
*Effects of Cooperation and Representation
on Measures of Intergroup Bias*

Measure	No cooperation		Cooperation	
	Two groups	One group	Two groups	One group
Like				
In-group	5.38	5.12	5.62	5.53
Out-group	4.48	4.75	5.19	5.27
Bias	0.90	0.36	0.43	0.26
Honest				
In-group	5.89	5.49	6.17	5.97
Out-group	4.98	5.15	6.08	6.01
Bias	0.90	0.35	0.08	-0.04
Cooperative				
In-group	5.93	5.44	5.88	5.96
Out-group	5.09	5.13	5.66	5.87
Bias	0.84	0.31	0.22	0.09
Similar to me				
In-group	4.64	4.23	4.85	4.92
Out-group	3.95	4.05	4.41	4.74
Bias	0.69	0.18	0.44	0.18
Average				
Index _{in-group}	5.46	5.07	5.63	5.59
Index _{out-group}	4.63	4.77	5.33	5.47
Bias	0.83	0.30	0.30	0.12

jects receiving the one-group manipulation selected the one-group alternative more frequently, 39.7% versus 15.8%, $F(1, 36) = 21.38, p < .001$, and rated (on scales ranging from 1 to 7) the aggregate more strongly as one group than did subjects receiving the two-group manipulation, $F(1, 36) = 9.84, p < .003$.

Second, to determine whether the representation manipulation affected intergroup bias in the absence of cooperative interaction, we conducted a 2 (representation) $\times$ 2 (sex) $\times$ 2 (in-group, out-group) multivariate analysis of variance (MANOVA), with repeated measures on the in-group, out-group variable, on the four evaluative ratings within only the no-cooperation condition. The results of this analysis revealed the expected Representation $\times$ In-group, Out-group interaction, $F(4, 33) = 2.55, p = .058$, each of the four univariate effects reached statistical significance at least at the .02 probability level. In this and subsequent analyses of bias, there were no effects involving subjects' sex. Examination of the evaluative indexes composing the bias measure, calculated to illustrate the pattern of the multivariate effects (see Table 2), indicated that bias (i.e., Index_{IN} - Index_{OUT}) in the no-cooperation/one-group condition (.30) was lower than in the no-cooperation/two-groups condition (.83). Although these findings were consistent with expectations, closer examination revealed that bias was reduced here because out-group members were evaluated more favorably (.14) in the one-group condition (Index_{OUT} = 4.77) than in the two-groups condition (Index_{OUT} = 4.63), which was

¹ Comparable univariate analyses involving these indexes yielded similar findings to those reported at the multivariate level.

expected, and also because in-group members were regarded less favorably ($-.39$) in the one-group condition ($\text{Index}_{\text{IN}} = 5.07$) than in the two-groups condition ($\text{Index}_{\text{IN}} = 5.46$). Thus, the between-groups comparisons suggest that there is a causal relationship between the representation manipulation and intergroup bias in the absence of cooperative interaction.²

A second hypothesis was that if cooperation influences members' representations of the aggregate, then within only the two-groups condition, cooperation would transform members' representations from two groups to one group and consequently would reduce bias. Within only the two-groups conditions ($n = 40$ groups), the results of a series of planned comparisons between the no-cooperation and cooperation conditions supported these expectations.

First, 2 (cooperation) $\times 2$ (sex) ANOVAs conducted on measures of subjects' representations within only the two-groups conditions revealed the expected main effects for cooperation. For example, the average percentage of subjects selecting the one-group representation was greater for the cooperation (48.3%) than for the no-cooperation (15.8%) condition, $F(1, 36) = 22.7$, $p = .001$. Similarly, the extent (as measured on a scale ranging from 1 to 7) to which the aggregate felt that one group was greater in the cooperation condition ($M = 4.44$) than in the no-cooperation condition ($M = 2.72$), $F(1, 36) = 38.03$, $p < .001$. Measures of the two-groups representation corroborated this pattern of findings.

Second, in terms of whether cooperation reduces bias, the results of a 2 (cooperation) $\times 2$ (sex) $\times 2$ (in-group, out-group) MANOVA with repeated measures on the in-group/out-group variable, which was performed on the four evaluative ratings within just the two-groups conditions yielded the expected Cooperation $\times$ In-group/Out-group interaction, $F(4, 33) = 4.82$, $p < .004$. The evaluative indexes for in-group and out-group members (see Table 2), within just the two-groups conditions, indicated that indeed, intergroup bias (i.e., $\text{Index}_{\text{IN}} - \text{Index}_{\text{OUT}}$) was lower in the cooperation/two-groups condition ($.30$) than the no-cooperation/two-groups condition ($.83$). Furthermore, the pattern of change responsible for lower bias that was due to intergroup cooperation here was consistent with our expectations. That is, although in-group members were regarded slightly more favorably ($.17$) in the cooperation ($\text{Index}_{\text{IN}} = 5.63$) versus no-cooperation ($\text{Index}_{\text{IN}} = 5.46$) conditions, the change for out-group members ($.70$) was even greater ($\text{Index}_{\text{OUT}} = 5.33$ vs. 4.63 , respectively). Given the pattern of these means, the Cooperation $\times$ In-group/Out-group interaction effect within just these two-groups representation conditions (mentioned previously) indicated that change sustained by out-group members was, as expected, reliably greater than the change for in-group members.³

The between-treatments analyses were strongly supportive of the predictions derived from the model that proposes that the relationship between intergroup cooperation and reduced intergroup bias is mediated by transformations in members' representations of the aggregate from two groups to one group.

Multiple Regression Mediation Approach

Further evidence more directly supportive of the proposed model is offered by the multiple regression mediation ap-

proach suggested by Baron and Kenny (1986) and Judd and Kenny (1981). The multiple regression mediation analysis (a form of path analysis) requires that a series of regression equations be estimated. In one equation, the dependent variable, for example, bias ($\text{Index}_{\text{IN}} - \text{Index}_{\text{OUT}}$) regressed on the independent variables (e.g., cooperation and representation) should demonstrate that the independent variable(s) affect the dependent variable; this of course would be similar to main effects for cooperation and representation on bias in an overall ANOVA between-treatments approach. In other equations, each of the potential mediators (i.e., the extent, ranging from 1 to 7, to which the aggregate felt like one group, two groups, or separate individuals and the degree of perceived cooperativeness or competitiveness during the contact period) that are regressed on the independent variables should indicate that the independent variable(s) also affect one or more of the potential mediators. These effects were reported earlier in the between-treatments analyses. Finally, and most advantageously, another equation in which the dependent variable (e.g., bias) is regressed on the independent variables together with the set of potential mediators should indicate that at least one of the potential mediators affects the dependent variable over and above the influence of the other variables. Also, in this last equation, the effect of the independent variable(s) should be weaker than in the initial equation because if there was perfect mediation, the independent variable(s) would have no effect beyond the association between the mediator and the dependent variable.

This multiple regression mediation approach was particularly useful for the current research because it permitted an estimate of any independent effects of the different representations of the aggregate; thus it more directly examined portions of the proposed model than would be possible with the between-treatments approach alone. For example, it is possible that, contrary to the details of our theoretical view, the only representation that mediates intergroup attitudes is the degree to which the memberships categorize themselves as two groups. Once the categorized representation is degraded, bias may be

² Furthermore, correlational analyses within these no-cooperation conditions involving subjects' representational ratings (1–7) further suggest that the one-group representation related to intergroup bias as expected. The more strongly subjects indicated that the aggregate felt like one group, the lower the bias, $r(38) = -.5134$, $p < .001$, and the higher the rating of out-group members, $r(38) = .5070$, $p < .001$. The ratings of the one-group representation did not relate to the evaluation of in-group members, $r(38) = .0336$, *ns*. In general, then, the patterns of relationships between representation and bias in the no-cooperation condition supported our hypothesis that representations of the aggregate, in the absence of cooperation, could influence intergroup bias.

³ Furthermore, correlational analyses involving subjects' representational ratings, ranging from *not at all* (1) to *very much* (7), and intergroup evaluations again suggested support for the proposed relationship between a one-group representation and intergroup bias. The more strongly subjects rated the aggregate as one group, the lower the bias, $r(38) = -.6085$, $p < .001$, and the higher the rating of out-group members, $r(38) = .7776$, $p < .001$. Also, ratings of the one-group representation did not relate to the evaluation of in-group members, $r(38) = .2292$, $p < .15$.

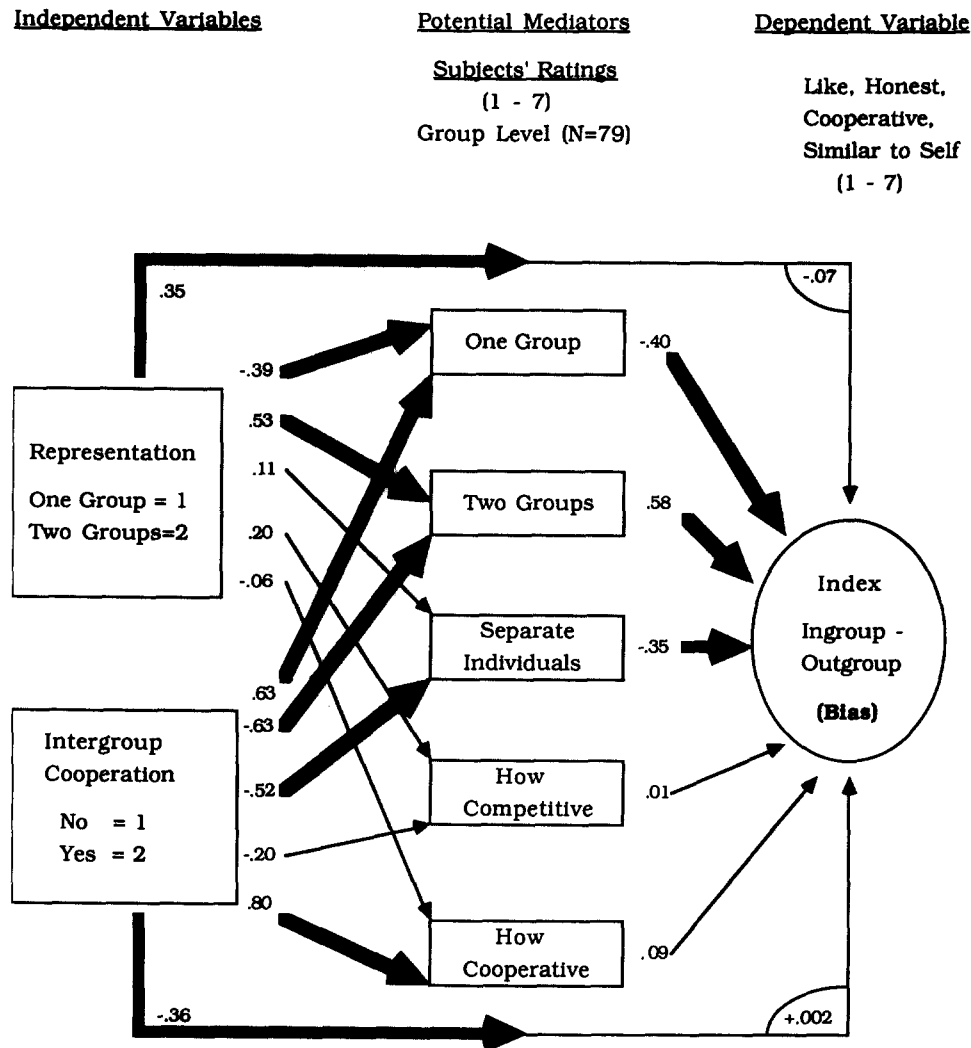


Figure 1. The mediation analysis of the effects of representation and cooperation on intergroup bias. (Bold arrows indicate that $p < .05$.)

reduced regardless of whether the two-groups categorization is replaced by some other representation (e.g., as one group or as separate individuals). Our earlier work, however, indicated that different residual representations reduced bias in different ways (Gaertner et al., 1989). Therefore, because we expected the ratings of the one-group and separate-individuals representations to relate differently to feelings toward in-group and out-group members, we conducted a separate mediation analysis for each of three relevant dependent variables—bias ($\text{Index}_{\text{IN-OUT}}$, see Figure 1), Index_{IN} , and $\text{Index}_{\text{OUT}}$ —to more fully examine the processes by which different representations might reduce bias. Statistically reliable relationships beyond the .05 level are indicated by the bolder arrows connecting these variables.

As indicated in Figure 1, when entered by themselves, the independent variables of representation and cooperation (see left column of Figure 1) had standardized beta weights of .35 ($p < .001$) and $-.36$ ($p < .005$) with the degree of bias (In-

$\text{dex}_{\text{IN-OUT}}$, represented in the oval to the right). These results paralleled the main effects for these variables obtained in an overall between-treatments analysis.⁴ Also, these independent

⁴ An overall 2 (representation) $\times$ 2 (cooperation) $\times$ 2 (sex) $\times$ 2 (in-group, out-group) multivariate analysis of variance (MANOVA) with repeated measures on the in-group/out-group variable was performed on the four evaluative ratings. This overall analysis yielded both a Representation $\times$ In-group/Out-group interaction, $F(4, 68) = 3.25$, $p < .017$, as well as a Cooperation $\times$ In-group/Out-group, $F(4, 68) = 7.26$, $p < .001$, interaction. Bias ($\text{Index}_{\text{IN}} - \text{Index}_{\text{OUT}}$) was lower in the one-group condition ($5.32 - 5.11 = .21$) than in the two-groups condition ($5.54 - 4.98 = .56$); also, bias was lower in the cooperation condition ($5.61 - 5.40 = .21$) than in the no-cooperation condition ($5.26 - 4.70 = .56$). Examination of these means revealed that the reduced bias in the one-group condition, in relation to the two-groups condition, was attributable both to a more positive attitude toward out-group members (.13) as well as to a less positive attitude toward in-group members

variables each relate to several of the potential mediators (see center column). For example, cooperation is related to the ratings of one group ($\beta = .63$), two groups ($\beta = -.63$), and also to subjects' ratings of cooperation ($\beta = .80$). Also, in the last equation, when the manipulations of representation and cooperation were entered into the regression equation together with subjects' ratings of one group, two groups, separate individuals, cooperation, and competition (adjusted $R^2 = .434$), each of the ratings of subjects' representations of the aggregate were predictive of intergroup bias. Specifically, the betas for one group, two groups, and separate individuals with bias were $-.40$, $.58$ and $-.35$, respectively. Consistent with our earlier work, both the one-group and separate-individuals representations were negatively associated with intergroup bias, in contrast to the positive association for the two-groups representation. Also, as we would expect if mediation had been occurring, when all variables were entered together into the last equation, the independent variables of representation ($\beta = -.07$) and cooperation ($\beta = .002$) no longer related reliably to intergroup bias.

Also note that in the current experiment, the ratings of cooperation and competition did not appear to relate independently to bias once the representation ratings were controlled. Nevertheless, their effects might have been suppressed by one or more of the other variables because of potential multicollinearity. As would be expected by our hypothesis, the ratings (1–7) of cooperation and one group were reliably related $r(77) = .685$, $p < .001$. Supplemental analyses for the last equation suggested that when bias was regressed on the full set of other variables, the betas for the one-group and cooperation ratings were correlated negatively but weakly, $r(71) = -.277$, $p < .05$.⁵ However, the tolerance of the one-group and cooperative ratings (tolerance = $.249$ and $.280$, respectively) suggested that their beta weights were not so influenced by the set of other variables as to be wholly determined by them.

The additional mediation analyses for the ratings of in-group members only indicated that in the last equation, the ratings of the separate-individuals representation ($\beta = -.43$) and two-groups representation ($\beta = .53$) related to in-group ratings (adjusted $R^2 = .307$). The less subjects felt like separate individuals and the more they felt like two groups, the more positively they evaluated original in-group members. However, in the last equation in the mediation analyses involving only the ratings of

out-group members (adjusted $R^2 = .533$), only the ratings of one group ($\beta = .58$) related reliably to the out-group ratings. As expected, the more subjects felt like they were one group, the more positive were their ratings of original out-group members.

In terms of the full design of the experiment, the regression analyses indicated that the manipulations of representation and cooperation each influenced members' representations of the aggregate, which in turn were the major factors that influenced attitudes toward in-group and out-group members. However, a more specific test of the proposed model involved the effects of cooperation when it was the sole potential contributor to variation in members' representations. Therefore, another set of multiple regression mediation analyses were conducted involving only subjects (i.e., groups) in the two-groups conditions ($n = 40$), so that the effects of cooperation alone could be assessed. Thus, these analyses excluded the representation manipulation and were again conducted separately for the three relevant dependent variables: bias ($\text{Index}_{\text{IN-OUT}}$), Index_{IN} , and $\text{Index}_{\text{OUT}}$.⁶ When only the independent variable of cooperation was considered, the results—despite the reduction in sample size—continued to demonstrate the importance of subjects' representations of the aggregate as mediators of the causal relation between intergroup cooperation and bias.

As indicated by Figure 2, when bias was regressed (in the last equation) on all of the other variables (adjusted $R^2 = .48$), only the two-groups ($\beta = .72$) and the separate-individuals ($\beta = -.44$) representation ratings influenced bias. The more the aggregate felt like two groups and the less it felt like separate individuals, the higher was the degree of bias. Thus, when the representation manipulation was omitted, the ratings of the one-group representation ($\beta = -.13$) did not relate independently to bias (although the simple correlation between the one-group representation ratings and bias was reliable, $r(38) = -.6085$, $p < .001$). However, the additional regression analyses involving the separate components of bias, namely, the evaluations of in-group and out-group members separately, more spe-

($-.22$). However, the reduced bias associated with the cooperation condition, in relation to the no-cooperation condition was due primarily to the greater enhancement of attitudes toward out-group members (.70) than toward in-group members (.30). Given the pattern of means involved in these comparisons, the Cooperation $\times$ In-group/Out-group MANOVA interaction reported in this note, indicated that this positive change toward out-group members was reliably more substantial than the change toward in-group members. Although not directly related to the framework that initiated the research, this MANOVA also yielded a main effect (across both in-group and out-group ratings) for cooperation, $F(4, 68) = 11.44$, $p < .001$. Overall, the evaluative ratings were higher for both in-group and out-group members in the cooperation ($M = 5.51$) than in the no-cooperation (4.98) condition. No other effects were obtained.

⁵ The notion of a correlation between betas can be somewhat confusing because it is not obvious how we could obtain a correlation involving only a single pair of values. The meaning of this statistic should be understood in terms of a correlation between an infinitely large set of estimated beta pairs, each based on one of an infinitely large number of (fixed-effects) replications of the present experiment. Each replication generates a somewhat different value for the dependent variable (i.e., bias) and thereby results in different estimates for these two betas. The correlation between these infinite number of beta pairs can be estimated (see Finn, 1974, page 100); it is this estimate that is reported.

⁶ For just the two-groups conditions, we hypothesized that cooperation would reduce bias primarily by changing subjects' representations from the baseline of two groups to one group. Thus, because the one-group representation was theoretically presumed to decrease the distance between the self and out-group members, we anticipated that reduced bias in the two-groups conditions would be due more strongly to changes in the evaluations of out-group members than to changes involving in-group members. Consistent with this reasoning, in the two-groups conditions, reductions in bias ($\text{Index}_{\text{IN-OUT}}$) were correlated more highly, $t(37) = 3.03$, $p < .01$, with more positive ratings of out-group members, $r(38) = -.72$, $p < .001$, than they were with less positive ratings of in-group members, $r(38) = .37$, $p < .009$.

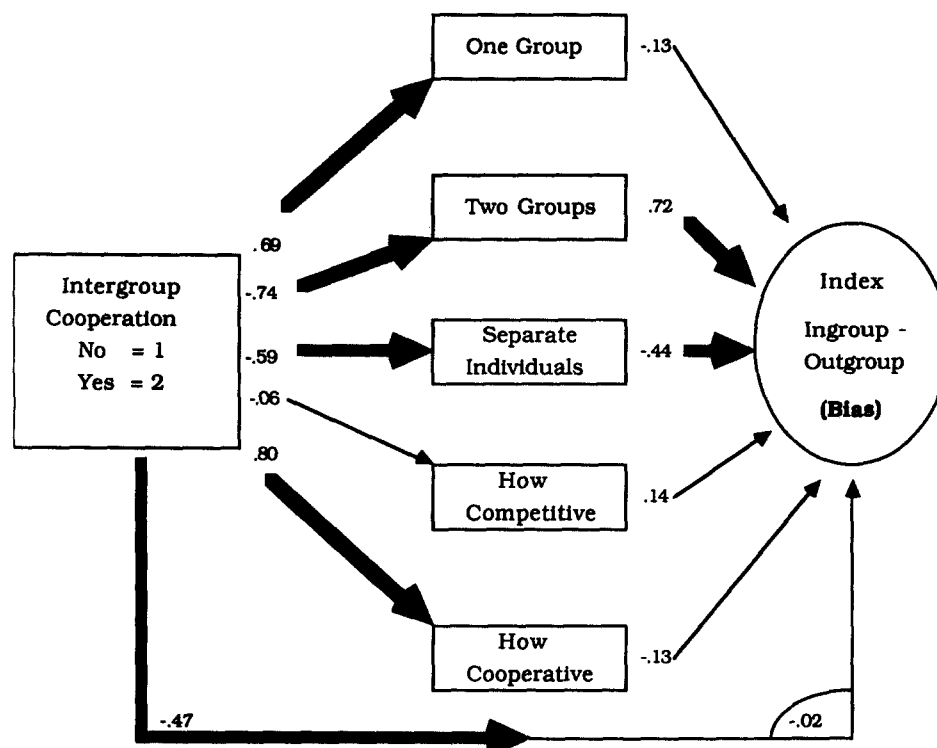


Figure 2. The mediation analysis (for the two-groups conditions only) of the effects of cooperation on intergroup bias. ($n = 40$. Bold arrows indicate $p < .05$.)

cifically clarify the special role of the one-group representation as it mediates the relationship between intergroup cooperation and reduced bias.

When the evaluations of former in-group members were considered separately and regressed on all of the other variables in the last equation (adjusted $R^2 = .10$), only the two-groups ($\beta = .61$) and the separate-individuals ($\beta = -.54$) representation ratings were predictors of the evaluations of in-group members. Thus, attitudes toward in-group members are not influenced primarily by the one-group representation.

The separate regression analyses involving the evaluations of out-group members, however, revealed (see Figure 3) that when the out-group evaluations were regressed (in the last equation) on the full set of other variables (adjusted $R^2 = .58$), only the ratings of the one-group representation ($\beta = .47$) reliably and independently influenced these ratings of out-group members. As expected, the greater the extent to which the aggregate felt like one group, the more positive were the evaluations of out-group members. Thus, although the one-group representation did not independently affect bias or the ratings of in-group members, it was the only reliable determinant of the attitudes toward out-group members.

Discussion

Overall, the results offer convergent support for the social categorization approach and the hypothesis that the causal relation between intergroup cooperation and reduced bias is me-

diated by transformations in members' cognitive representations of the aggregate. Specifically, cooperation degraded the two-groups representation and induced the memberships to recategorize themselves primarily as one larger group. Also, the more strongly the aggregate was regarded as one group, the more highly out-group members were evaluated, a consequence that contributed substantially to the reduction of intergroup bias.

The multiple regression mediation analyses on the full experimental design (see Figure 1), in which cooperation and the additional features constituting the representation manipulation could both contribute to the induction of a one-group representation, suggested that the one-group representation ratings could relate independently to changes in bias, as well as to the evaluations of out-group members. These circumstances presented by the full design may be structurally comparable to some natural settings in which intergroup cooperation is accompanied by other features, which according to the elaborated version of contact hypothesis (Allport, 1954; Brewer & Miller, 1984; Cook, 1978; Pettigrew, 1971; Stephan, 1987), have the potential for influencing the results of the contact experience. Perhaps some of these features (e.g., equal-status contact, the opportunity for more intimate encounters, and equalitarian norms) affect the quality of the contact experience because they further contribute to transformations of members' categorized representations (Brewer & Miller). From our perspective, some of these features may also further encourage the development of a one-group representation.

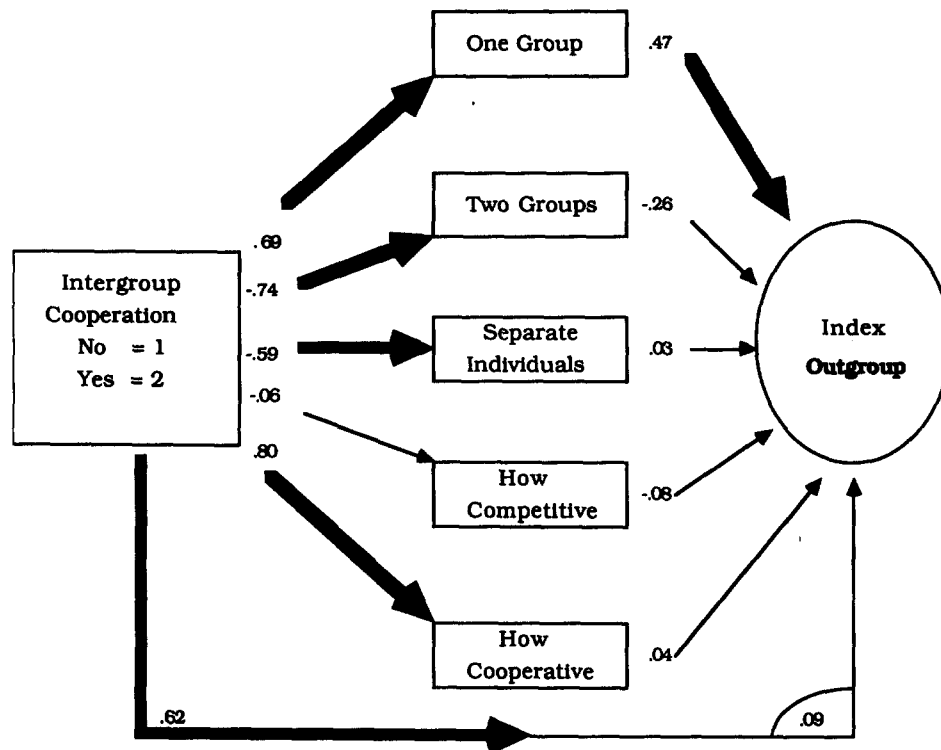


Figure 3. The mediation analysis (for the two-groups conditions only) of the effects of cooperation on the evaluations of out-group members. ($n = 40$. Bold arrows indicate $p < .05$.)

In the regression analyses when the representation manipulation was omitted (i.e., considering only the two-groups-representation conditions; see Figures 2 & 3), the one-group representation did not independently affect bias per se, but rather exerted its influence primarily by enhancing the evaluations of out-group members. This pattern, however, is consistent with the proposed model. To the extent that cooperation reduced bias by transforming members' representations from two groups to one group, bias was reduced primarily because attitudes toward former out-group members became more positive. With the revised one-group representation induced by cooperation, former out-group members were regarded as generally more likable, cooperative, honest, and similar to the self. These results are supportive of Turner's (1985) theory of self-categorization and Brewer's (1979) conclusion that group formation primarily brings in-group members closer to the self.

The presence of cooperation distinctly altered subjects' representations of the memberships and also the pattern of in-group and out-group attitudes. With cooperation, the one-group representation dominated subjects' impressions in the one-group (66.7%) and the two-groups (48.3%) conditions (see Table 1); whereas without cooperation, subjects selected this representation only 39.7% and 15.8% of the time, respectively.⁷ In relation to the set of other features intended to influence subjects' representations, cooperation sharply increased the percentages of subjects selecting the one-group alternative in both the one-group and especially in the two-groups conditions. Also, intergroup bias was lower with cooperation. This finding also was

somewhat more noteworthy in the two-groups conditions. Furthermore, the reduced bias associated with cooperation was attributable primarily to the development of more positive attitudes toward former out-group members. Overall, the pattern of these findings is supportive of the proposed model.

Unexpectedly, cooperation also decreased the extent to which subjects endorsed the Separate-Individuals representation, albeit from weak to barely negligible (see Table 1 and Figures 1 & 2). Thus, although subjects did not frequently choose the separate-individuals representation to characterize themselves (i.e., it was only selected 7.5% of the time overall, and subjects did not rate it strongly on the scale ranging from *not at all* (1) to *very much* (7), $M = 2.16$), cooperation, nevertheless, still further reduced the strength of this impression. In retrospect, this change for the separate-individuals representation was reasonable, if indeed cooperation emphasizes one's state of belonging to the superordinate entity. This finding is important to our theoretical framework because the mediation analyses indi-

⁷ Despite the weakness of the one-group representation when it was induced by factors other than cooperation, the reduced bias was due partially to the enhanced attitudes toward out-group members as well as to less positive attitudes toward in-group members. The reason that attitudes toward in-group members were less positive in the one-group condition than in the two-groups condition in the absence of cooperation may be related to Wilder's (1986) suggestion that the salience of a specific outgroup (as would be the case in our two-groups condition) can increase favorable regard for in-group members.

cated that higher ratings of separate individuals was associated with lower levels of intergroup bias and with decreased attraction to former in-group members. These latter findings are consistent with our previous work (Gaertner et al., 1989), with Brewer's (1979) analysis, and with Turner's (1985) self-categorization theory.⁸

In general, the results offer support for the model that initiated the research. Nevertheless, we do not believe that intergroup cooperation reduces bias solely by modifying members' representations of the aggregate. Statistically, the regression analyses, which suggested that there was no independent causal relation between the cooperation treatment variable (or ratings of cooperation) with bias, can be misleading. For example, to the extent that there may be collinearity between the ratings of cooperation and representation measures, the statistical power associated with these variables may be reduced. Discussing the potential problems caused by collinearity, Judd and Kenny (1981, p. 607) indicate, "We may conclude that a particular variable in a causal chain exerts no effect upon the next variable simply because of poor power."

Furthermore, within the experimental design, the variety of processes by which cooperation could have reduced bias might not have been fully represented. For example, in the current study, the effects of cooperation were compared with a control condition involving no cooperation, rather than with competition. Indeed, the design was intended to assess the effects of introducing intergroup cooperation per se, rather than the effects of removing intergroup competition. In Sherif et al.'s (1954) study, a series of intergroup cooperative encounters produced harmony among groups with a history of competitiveness and hostility toward the other. By contrast, in the current study, one cooperative episode reduced mild levels of largely pro-in-group bias produced primarily by social categorization per se without prior history of competitive or hostile relations between the groups. Worchel et al. (1978) showed that the degree of bias reduced by intergroup cooperation is dependent on the nature of the earlier relationship between groups (i.e., whether they were cooperative, competitive, or independent) as well as by the degree of success achieved by the cooperative venture. Therefore, the relative strength of the different processes by which cooperation can reduce bias may also vary differentially depending on the circumstances. Finally, the present study was designed primarily to examine the efficacy of the proposed theoretical model and not to test its effectiveness in relation to each of the other models. Therefore, for example, we did not include a success/failure manipulation, which would have more directly provided an opportunity to test reinforcement theory.

Although members' representations are probably not the only mediators of the relation between cooperation and intergroup bias, it is clear from this study that although the other major theories are not disproved, the other explanations would have difficulty explaining the reported findings without including members' representations in the causal chain. In particular, the lower bias obtained between the one-group condition and the two-groups condition without cooperation would be problematic for the other positions. During intergroup contact without cooperation in this study, there is little reason to believe

that subjects in the one-group condition in relation to the two-groups condition engaged in behavior that created a state of imbalance to be rectified, gained more knowledge about the other group or its members, or experienced greater levels of positive reinforcement. Also, the degree to which members' conceived of themselves as separate individuals did not differ for these conditions; therefore it is unlikely that subjects had more individuated or personalized impressions of out-group members in the one-group than in the two-groups conditions. However, these conditions differed in terms of the extent to which the manipulations transformed members' representations from two groups to one group. In addition, these results weaken the possibility that in the cooperation condition, reduced bias (the final dependent variable in the regression analysis) preceded the transformations in members' representations (the proposed mediators). Indeed, in the no-cooperation condition, in which the direction of causality was more certain because of the elimination of interaction, changes in members' representations most plausibly preceded changes in bias.

Intergroup cooperation increased the extent to which subjects adopted a one-group representation, and intergroup cooperation also decreased intergroup bias. However, which features of the cooperation treatment were most responsible or necessary and sufficient to produce these effects? Compared with the no-cooperation condition, participants in the cooperation condition interacted with one another, worked together toward a common goal, and shared the same fate regarding the outcome of the joint venture. Certainly, each of these features was involved when Sherif and Sherif's (1969) groups of summer campers cooperated in their search for leaks to the camp's water supply and during their attempts to get the food truck started. Similarly, each of these features is involved in cooperative team-learning interventions which have positive effects on intergroup behavior (Johnson et al., 1983; Slavin, 1985).

Also, intergroup cooperation that permits each group to work separately but to have equally important and complementary roles toward achieving a superordinate objective may be especially effective at reducing bias (Brown & Wade, 1987; Deschamps & Brown, 1983). For example, when group affiliation is meaningful to members' self-identities, intergroup cooperation that allows each group to make favorable and distinct

⁸ The regression analyses (see Figures 1 & 2), however, suggest that as cooperation decreases the salience of a separate-individuals representation, it contributes to an increase in intergroup bias. Although these findings were unexpected, they are consistent with a theoretical framework that emphasizes the mediating role played by members' representations of the aggregate. Paradoxically, to the extent that intergroup cooperation induces a one-group representation, it can contribute to reduced bias, whereas, to the extent that cooperation decreases the separate-individuals representation, it can increase intergroup bias. This result suggests that the specific effect of cooperation on bias depends on the pattern of representations induced by cooperation and, more specifically, by relative strength of the different residual representations. The findings indicate that intergroup cooperation degrades the two-group representation and most strongly increases the salience of a one-group representation. Consequently, cooperation generally reduces bias.

tive intergroup comparisons may permit the memberships to develop mutual respect for one another without threatening their own positive group identities (Deschamps & Brown, 1983; see also Hewstone & Brown, 1986). This position is not necessarily incompatible with our own. Previously, we suggested that the acceptance of a common in-group identity may not necessarily require subgroups to forsake their earlier categorizations entirely (Gaertner et al., 1989). Within the context of a more inclusive superordinate group representation, the salience of these subgroup identities may enable the enhanced appreciation for out-group members to generalize beyond those directly involved in the cooperative effort. We would expect, however, that the extent of these changes would be mediated more by the strength of the superordinate group representation than by the extent to which the subgroups were salient.

Although researchers are learning more about the relationship between intergroup cooperation and intergroup bias, a major obstacle faces those wishing to apply this knowledge. As noted by Worchel (1979, 1986), it is often difficult to inspire cooperation among conflicting groups. However, it is our hope that we could identify alternative strategies for reducing intergroup bias if we could more precisely understand how cooperation reduces bias, which specific aspects of cooperative interaction are most critical, and why the other features specified by the elaborated contact hypothesis are beneficial. The current work, for example, suggests that there may be some value in trying to transform members' categorized representations of one another. In some contexts, this may be more feasible than directly inducing intergroup cooperation or orchestrating equal-status contact. Modifying members' representations even slightly may encourage the initiation of more constructive and cooperative intergroup relations.

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